

Business Requirement Document

Checkout & Payment Module

ShopSphere E-Commerce Platform — Platform Modernization Program (FY2026)

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Author	Arjun Rao, Business Analyst — Payments & Finance
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1. Document Information

Program	ShopSphere Platform Modernization Program (FY2026)
Module	Checkout & Payment
Prepared For	Rajesh Mehta, VP of E-Commerce (Executive Sponsor)
Distribution	Product, Business Analysis, IT/Architecture, QA, Design, Engineering/Infra, Risk & Compliance

2. Executive Summary

The Checkout & Payment module is the revenue-critical core of the ShopSphere E-Commerce Platform, processing the final steps of the customer purchase journey across web and mobile applications for approximately 2 million monthly active users in India, UAE, and Southeast Asia. This Business Requirement Document defines the requirements for modernizing the checkout and payment experience as part of the FY2026 ShopSphere Platform Modernization Program, with the objective of reducing checkout abandonment, improving payment authorization success rates, strengthening fraud and chargeback controls, and ensuring full PCI-DSS Level 1 compliance across all supported payment instruments. The scope covers guest and registered-user checkout flows, address and shipping management, payment method selection and processing via the PayEdge Payment Gateway, real-time fraud screening through the Signifyd Fraud Detection Service, region-specific tax computation (GST in India, VAT in UAE), and order confirmation with e-invoice generation synchronized to SAP S/4HANA. This document consolidates business, functional, and non-functional requirements, along with user stories, acceptance criteria, business rules, risks, and success metrics, to serve as the authoritative reference for design, development, testing, and sign-off by all program stakeholders.

3. Business Objective

The primary business objective of the Checkout & Payment module is to deliver a fast, secure, and frictionless purchase completion experience that maximizes checkout conversion while minimizing payment failures, fraudulent transactions, and chargebacks. ShopSphere Technologies Pvt. Ltd. aims to increase overall Gross Merchandise Value (GMV) by reducing cart-to-order drop-off, support diverse regional payment preferences (cards, UPI, net banking, wallets, and Cash on Delivery), and comply with evolving regulatory requirements across India, UAE, and Southeast Asian markets. The modernization effort also seeks to reduce operational costs associated with manual fraud review and payment reconciliation by introducing automated, real-time risk scoring and straight-through processing wherever feasible, while maintaining a customer experience that remains within the two-second response threshold defined by the platform's non-functional baseline.

4. Project Scope

In Scope

- Guest checkout and registered-user checkout workflows for web and mobile app.
- Shipping and billing address capture, validation, and management (add, edit, delete, set default).
- Shipping method selection, including estimated delivery windows and shipping cost computation.

- Payment method selection and processing: credit/debit card, UPI, net banking, digital wallets, and Cash on Delivery (COD).
- Order confirmation, order summary display, and e-invoice generation.
- Real-time fraud screening and risk scoring prior to order confirmation.
- Region-specific tax calculation: GST for India, VAT for UAE, and applicable indirect taxes for Southeast Asian markets.

Out of Scope

- Subscription and recurring billing arrangements.
- Split payment across multiple payment instruments within a single order.
- Third-party marketplace seller payout disbursement and settlement processing.

5. Stakeholders

Name	Role	Department	Responsibility
Rajesh Mehta	VP of E-Commerce	Business	Executive Sponsor
Ananya Sharma	Product Owner, ShopSphere Platform	Product	Prioritization & sign-off
Arjun Rao	Business Analyst, Payments & Finance	Business Analysis	Requirements owner
Sarah Chen	Enterprise IT Architect	IT/Architecture	Technical feasibility
Michael Torres	QA Lead	Quality Assurance	Test strategy & sign-off
Priya Nair	UX Lead	Design	User experience
David Kim	DevOps Lead	Engineering/Infra	Deployment & monitoring
Fatima Al-Sayed	Compliance & Security Officer	Risk & Compliance	Regulatory compliance

6. Business Requirements

ID	Business Requirement
BR-001	The platform must allow customers to complete a purchase without mandatory account registration, offering a guest checkout path to reduce friction and abandonment.
BR-002	The platform must support the full range of regionally relevant payment methods, including credit/debit cards, UPI, net banking, digital wallets, and Cash on Delivery, to maximize payment acceptance across India, UAE, and Southeast Asia.
BR-003	The platform must screen every transaction for fraud risk in real time to reduce chargeback losses and protect the business from payment fraud.
BR-004	The platform must calculate and display accurate, region-specific taxes (GST in India, VAT in UAE) at checkout so that customers see the final payable amount before confirming payment.
BR-005	The platform must generate a compliant e-invoice for every completed order and post the transaction to SAP S/4HANA for financial reconciliation.
BR-006	The platform must ensure all payment processing complies with PCI-DSS Level 1 requirements to protect customer financial data and maintain acquiring bank relationships.
BR-007	The platform must provide customers with clear, real-time feedback on payment failures and enable a low-friction retry path to recover otherwise-lost sales.
BR-008	The platform must restrict Cash on Delivery availability based on defined eligibility rules (order value, delivery pincode, and customer risk profile) to limit COD-related losses.

7. Functional Requirements

Address & Shipping

- **FR-001** The system shall allow registered users to save, edit, delete, and designate a default shipping and billing address, with a maximum of 10 saved addresses per account.
- **FR-002** The system shall validate address fields (pincode/postal code, city, state/emirate, country) against a reference postal database and reject unserviceable pincodes at checkout entry.
- **FR-003** The system shall display available shipping methods with associated cost and estimated delivery date range, computed based on the delivery address, SKU weight/dimensions, and warehouse of fulfillment (NetSuite WMS / IMS).

- **FR-004** The system shall allow guest users to enter a one-time shipping and billing address without persisting it to a customer profile unless the guest converts to a registered account post-purchase.

Payment Processing

- **FR-005** The system shall present available payment methods (credit/debit card, UPI, net banking, wallet, COD) filtered by region, order value, and customer eligibility.
- **FR-006** The system shall route all card, UPI, net banking, and wallet transactions through the PayEdge Payment Gateway for authorization and capture, without storing raw card or bank credentials on ShopSphere infrastructure.
- **FR-007** The system shall support tokenized card storage via PayEdge for returning customers who opt to save a payment method, displaying only the last four digits and card network.
- **FR-008** The system shall trigger an OTP-based verification step for UPI and net banking transactions in accordance with regional two-factor authentication mandates.
- **FR-009** The system shall enforce a COD eligibility check at the payment step, evaluating order value ceiling, delivery pincode serviceability, and customer COD risk history before offering COD as an option.
- **FR-010** The system shall calculate applicable taxes (GST for India, VAT for UAE, regional equivalents for Southeast Asia) at both the line-item and order level and display an itemized tax breakdown prior to payment confirmation.

Fraud Checks

- **FR-011** The system shall submit each transaction to the Signifyd Fraud Detection Service in real time and receive a risk score prior to final order confirmation.
- **FR-012** The system shall automatically decline or hold for manual review any transaction scored above the configured high-risk threshold, routing held orders to a fraud review queue.

Order Confirmation

- **FR-013** The system shall generate a unique order number and an order confirmation page immediately upon successful payment authorization or COD order placement.
- **FR-014** The system shall generate a GST/VAT-compliant e-invoice in PDF format and make it available for download from the order confirmation page and order history.
- **FR-015** The system shall post confirmed order and payment data to SAP S/4HANA for order-to-cash processing within 60 seconds of order confirmation.
- **FR-016** The system shall trigger an order confirmation email via SendGrid and an order confirmation SMS via Twilio within 2 minutes of successful order placement.

8. Non-Functional Requirements

Performance

- **NFR-001** Checkout page load and payment initiation shall complete within 2 seconds at the 95th percentile under normal load.
- **NFR-002** Payment gateway authorization response shall be received and displayed to the customer within 4 seconds at the 95th percentile.
- **NFR-003** Tax calculation shall complete within 500 milliseconds per checkout page render.

Security

- **NFR-004** The module shall comply with PCI-DSS Level 1 requirements for all payment card data handling.
- **NFR-005** The module shall comply with OWASP Top 10 secure coding standards for all checkout and payment endpoints.
- **NFR-006** All PII and payment-related data shall be encrypted using AES-256 at rest and TLS 1.2+ in transit.

Availability

- **NFR-007** The checkout and payment services shall maintain 99.9% uptime for customer-facing operations, measured monthly.
- **NFR-008** Planned maintenance windows for payment services shall not exceed 2 hours per month and shall be scheduled outside peak traffic periods.

Scalability

- **NFR-009** The module shall scale to support up to 10x normal transaction volume during peak sale events such as Diwali and Black Friday without degradation in response time.
- **NFR-010** The payment queue and fraud-screening pipeline shall support horizontal auto-scaling based on transaction throughput.

Reliability

- **NFR-011** The module shall maintain a Recovery Point Objective (RPO) of 15 minutes for transaction data.
- **NFR-012** The module shall maintain a Recovery Time Objective (RTO) of 1 hour for checkout and payment services following a critical failure.

9. User Stories

- **US-001** As a first-time shopper, I want to check out as a guest without creating an account, so that I can complete my purchase quickly.
- **US-002** As a registered customer, I want to save multiple shipping addresses, so that I can select the correct one quickly on future orders.
- **US-003** As a customer in India, I want to pay using UPI, so that I can complete payment using my preferred method without entering card details.
- **US-004** As a price-sensitive customer, I want to choose Cash on Delivery when eligible, so that I can pay upon receiving my order.
- **US-005** As a customer, I want to see the exact tax amount (GST or VAT) applied to my order before I pay, so that I understand the total cost upfront.
- **US-006** As a customer, I want to receive immediate confirmation and an invoice after placing my order, so that I have proof of purchase.
- **US-007** As a customer whose payment fails, I want a clear error message and the option to retry with a different method, so that I don't lose my cart or have to restart checkout.
- **US-008** As a compliance officer, I want every transaction screened for fraud risk automatically, so that high-risk orders are caught before fulfillment.

10. Acceptance Criteria

ID	Related ID	Criteria	Expected Result
AC-001	FR-001	Given a registered user with 3 saved addresses, when they add a 4th address, then it is saved and selectable.	Address list updates without page reload; new address selectable at checkout.
AC-002	FR-002	Given a checkout in progress, when a customer enters an unserviceable pincode, then the system blocks progression.	Error message displayed identifying pincode as unserviceable.
AC-003	FR-006	Given a customer selects credit card payment, when they submit card details, then the transaction routes through PayEdge for authorization.	Authorization response received within 4 seconds; raw card data never persisted on ShopSphere servers.
AC-004	FR-009	Given an order value above the COD ceiling, when the customer reaches payment selection, then COD is not offered.	COD option hidden; alternate payment methods displayed.
AC-005	FR-010	Given a delivery address in Maharashtra, India, when checkout totals are computed, then GST is applied at the applicable rate.	Itemized GST breakdown (CGST/SGST) displayed on order summary.
AC-006	FR-011	Given a completed payment, when the Signifyd risk score exceeds the high-risk threshold, then the order is held.	Order status set to "Under Review"; order routed to fraud review queue; customer notified of delay.
AC-007	FR-014	Given a successfully confirmed order, when the confirmation page loads, then an e-invoice is available.	E-invoice PDF downloadable and compliant with GST/VAT invoice format.
AC-008	US-007	Given a declined card payment, when the failure is returned by PayEdge, then the customer is shown a retry option.	Cart contents preserved; customer can select an alternate payment method without re-entering address details.
AC-009	FR-016	Given a confirmed order, when order confirmation completes, then notifications are dispatched.	Email via SendGrid and SMS via Twilio received within 2 minutes.

11. Business Rules

- **RULE-001** Cash on Delivery shall not be offered for orders exceeding INR 50,000 (or regional equivalent) in order value.
- **RULE-002** A customer with more than 2 COD refusal incidents in the preceding 90 days shall be ineligible for COD on subsequent orders.
- **RULE-003** GST shall be computed as CGST + SGST for intra-state deliveries and IGST for inter-state deliveries within India.
- **RULE-004** VAT shall be applied at the UAE standard rate for all deliveries within the United Arab Emirates unless the SKU is designated zero-rated or exempt.
- **RULE-005** Any transaction receiving a Signifyd risk score above 85 (on a 0–100 scale) shall be automatically held for manual fraud review prior to fulfillment release.
- **RULE-006** Saved card tokens shall never be displayed with more than the last 4 digits and shall never expose the CVV.
- **RULE-007** An order shall not be released to the warehouse for fulfillment until payment is either successfully captured or, for COD, the order is confirmed and passes fraud screening.

12. Process Flow Description

Primary Flow — Checkout to Payment Confirmation:

1. Customer proceeds from cart to checkout as a guest or logged-in user.
2. Customer enters or selects a saved shipping address; the system validates pincode serviceability.
3. Customer selects a shipping method; the system computes shipping cost and estimated delivery date.
4. The system calculates order totals including region-specific tax (GST/VAT) and displays the final itemized summary.
5. Customer selects a payment method (card, UPI, net banking, wallet, or COD) from eligible options.
6. For electronic payments, the system routes the transaction to the PayEdge Payment Gateway for authorization, including OTP verification where applicable.
7. The system submits the transaction to the Signifyd Fraud Detection Service for real-time risk scoring in parallel with authorization.
8. Upon successful authorization and an acceptable fraud score, the system captures payment, generates an order number, and creates the e-invoice.
9. The system posts the order and payment record to SAP S/4HANA and updates order history in Salesforce CRM.
10. The system triggers order confirmation notifications via SendGrid (email) and Twilio (SMS), and the customer is shown the order confirmation page.

Alternate Flow — Payment Failure and Retry:

1. Customer submits payment details and the transaction is routed to PayEdge for authorization.
2. PayEdge declines the transaction (e.g., insufficient funds, bank decline, invalid OTP).
3. The system displays a clear, actionable failure message and preserves the cart and shipping selections.
4. Customer is offered the option to retry the same payment method or select an alternate method.
5. If retries are exhausted (3 consecutive failures), the system suggests COD (if eligible) or saving the cart for later completion.
6. Failed transaction attempts are logged and made available to fraud and support teams for pattern analysis.

13. Assumptions

- The PayEdge Payment Gateway maintains active acquiring bank relationships across India, UAE, and Southeast Asian markets.
- Customers have valid, active mobile numbers for OTP-based verification of UPI and net banking transactions.
- Regional tax rates (GST slabs, UAE VAT rate) are maintained and updated centrally in the tax engine and do not require per-transaction manual configuration.
- The Signifyd Fraud Detection Service integration provides risk scores within the required latency for real-time checkout decisions.
- SAP S/4HANA is available and reachable for order posting within the defined 60-second SLA window.
- Existing customer and address data migrated from legacy systems is accurate and complete.

14. Constraints

- All payment card data handling must remain within PCI-DSS Level 1 scope boundaries; no raw card data may be stored on ShopSphere-owned infrastructure.
- COD is not available in all serviceable pincodes due to logistics partner limitations.
- Tax computation logic must accommodate differing rules across India, UAE, and Southeast Asia within a single checkout engine.
- Peak sale events (e.g., Diwali, Black Friday) impose a hard scalability ceiling requiring infrastructure pre-provisioning.
- Third-party gateway and fraud service SLAs are outside ShopSphere's direct control and constrain end-to-end checkout latency.

15. Dependencies

- PayEdge Payment Gateway for all electronic payment authorization and capture.
- Signifyd Fraud Detection Service for real-time transaction risk scoring.
- SAP S/4HANA for order-to-cash posting and invoice reconciliation.
- SendGrid Email API and Twilio SMS Gateway for order confirmation notifications.
- NetSuite WMS and internal Inventory Management System (IMS) for shipping cost and delivery estimate computation.
- Salesforce CRM for order history and Customer 360 synchronization.

16. Risks

Risk ID	Description	Likelihood	Impact	Mitigation
RISK-001	PayEdge Payment Gateway outage during peak sale events causing checkout failures.	Medium	High	Implement gateway failover with a secondary payment processor and circuit-breaker pattern.
RISK-002	Increased chargebacks due to insufficient fraud screening coverage.	Medium	High	Tune Signifyd risk thresholds continuously and review declined/held transaction patterns monthly.
RISK-003	Incorrect GST/VAT computation leading to regulatory penalties or customer disputes.	Low	High	Centralize tax logic in a validated tax engine with quarterly compliance audits by Fatima Al-Sayed's team.
RISK-004	High COD refusal rates increasing reverse logistics costs.	High	Medium	Enforce COD eligibility rules (RULE-001, RULE-002) and monitor refusal rates by pincode.
RISK-005	Checkout page performance degradation during 10x peak traffic events.	Medium	High	Load-test checkout and payment services quarterly; enable auto-scaling ahead of known sale dates.
RISK-006	PCI-DSS non-compliance due to inadvertent storage of sensitive card data in logs.	Low	High	Implement automated log scrubbing and periodic PCI-DSS audits.

17. Data Requirements

The Checkout & Payment module captures and stores order, address, payment, and tax-related data. Payment card data is tokenized via PayEdge and never persisted in raw form within ShopSphere systems, consistent with PCI-DSS Level 1 scope reduction. Personally Identifiable Information (PII) such as customer name, address, and contact details is encrypted at rest (AES-256) and in transit (TLS 1.2+), and is retained in accordance with regional data protection regulations and the company's 7-year financial record retention policy for order and invoice data.

Data Entity	Key Fields	Classification	Retention
Order	Order ID, SKU list, quantities, order total, tax breakdown, status	Business/Financial	7 years
Payment Transaction	Transaction ID, payment method, PayEdge token reference, authorization status	PCI/Financial (tokenized)	7 years
Shipping/Billing Address	Name, address lines, pincode, city, state/emirate, country, phone	PII	Duration of account + 3 years
Fraud Risk Record	Signifyd risk score, decision, review outcome	Confidential	3 years
E-Invoice	Invoice number, GST/VAT registration, itemized tax lines	Financial/Regulatory	7 years

18. Integration Requirements

System	Purpose	Direction	Data Exchanged
PayEdge Payment Gateway	Payment authorization and capture for cards, UPI, net banking, wallets	Bi-directional	Payment token, transaction amount, authorization status, decline codes
SAP S/4HANA	Order-to-cash posting and invoice reconciliation	Outbound	Order ID, order total, tax breakdown, payment status, invoice reference
Signifyd Fraud Detection Service	Real-time transaction risk scoring	Bi-directional	Order details, customer/device metadata, risk score, decision
SendGrid Email API	Order confirmation and payment notification emails	Outbound	Customer email, order summary, e-invoice link
Twilio SMS Gateway	Order confirmation SMS and OTP delivery	Outbound	Customer mobile number, order number, OTP code
Salesforce CRM	Order history sync for Customer 360 view	Outbound	Order ID, customer ID, order status, order value

19. Reporting Requirements

- Daily checkout funnel and conversion rate dashboard segmented by device and region.
- Payment method mix and authorization success/failure rate report by gateway and payment type.
- Fraud screening outcome report showing approved, declined, and held-for-review transaction volumes.
- COD eligibility, usage, and refusal rate report by pincode and region.
- Tax collected report (GST/VAT) by region for finance reconciliation with SAP S/4HANA.
- Chargeback and dispute trend report for the Compliance & Security team.

20. Success Metrics

KPI	Target
Checkout conversion rate (cart to order)	Increase from baseline by at least 8%
Payment authorization success rate	≥ 96%
Fraud/chargeback rate	≤ 0.3% of total transaction volume
Average checkout completion time	≤ 90 seconds end-to-end
COD refusal rate	≤ 8% of COD orders
E-invoice generation accuracy	100% GST/VAT compliance, zero audit findings

21. Future Enhancements

- Support for split payment across multiple instruments within a single order.
- Buy-now-pay-later (BNPL) and installment payment options.
- Subscription and recurring billing support for consumable product categories.
- AI-driven dynamic payment method ranking based on customer conversion likelihood.
- One-click checkout using biometric authentication on mobile app.

22. Glossary

Term	Definition
SKU	Stock Keeping Unit — a unique identifier for a sellable product variant.
PDP	Product Detail Page — the page displaying detailed information about a single product.
PLP	Product Listing Page — a page displaying a list of products, typically by category or search.
Cart Abandonment	When a customer adds items to a cart but does not complete the purchase.
COD	Cash on Delivery — a payment method where the customer pays at the time of delivery.
RMA	Return Merchandise Authorization — an approval process for returning purchased goods.
SLA	Service Level Agreement — a defined commitment for service performance or response time.

OTP	One-Time Password — a single-use code used for transaction or identity verification.
KYC	Know Your Customer — identity verification process used for compliance purposes.
GMV	Gross Merchandise Value — total value of merchandise sold through the platform.
ATP	Available to Promise — inventory logic determining if stock can be committed to an order.
GST	Goods and Services Tax — India's indirect tax applied on the supply of goods and services.
VAT	Value Added Tax — indirect tax applied in the UAE and other jurisdictions on goods and services.
PII	Personally Identifiable Information — data that can identify a specific individual.

23. Appendix

Reference Documents:

- BRD-EC-CRT-002 — Shopping Cart Module
- BRD-EC-ORD-005 — Order Management Module
- BRD-EC-CUS-001 — Customer Account & Registration Module
- ShopSphere Platform Modernization Program Charter, FY2026

Revision History:

Version	Date	Author	Change Description
0.1	12-Jan-2026	Arjun Rao	Initial draft circulated for stakeholder review.
0.5	28-Jan-2026	Arjun Rao	Incorporated feedback from Sarah Chen (feasibility) and Fatima Al-Sayed (compliance).
1.0	05-Feb-2026	Arjun Rao	Final version approved by Ananya Sharma and Rajesh Mehta.